Date :

8. Abstraction

8.1 What is abstraction?

Abstraction is a process of hiding the implementation details and showing only functionality to the user.

There are two ways to achieve abstraction in Java.

1. Abstract class (0 to 100%)
2. Interface (100%)

Abstract class in Java :

A class which is declared as abstract is known as an abstract class. It can be have abstract and non-abstract methods. It needs to be extended and its method implemented. It cannot be instantiated. An abstract class must be delared with an abstract keyword. It can have abstract and non-abstract methods. It cannot be instantiated. It can have constructors and static method also. It can have final methods which will force the subclass not to change the body of the method.

Example:

```
abstract class A{}
```

Abstrat Method in Java:

A abstract method is a method which is declared as abstract and does not have implementation.

Example:

Date :

```
abstract void printstatus(); // no method body and abstract.
```

Example of abstract class having abstract method

```
abstract class Bike{
  abstract void run();
  }
class Honda4 extends Bike{
 void run(){System.out.println("running safely");}
 public static void main(String args[]){
  Bike obj = new Honda4();
  obj.run();
  }
 }
```

8.2 Interface :

An interface in Java is a blueprint of a class. It has static constants. The inheritance in java is a mechanism to achieve abstraction. There can be only abstract methods in the Java interface, not method body. It is used to achieve abstraction and multiple inheritance in java.

Java interface also represents the Is-A relationship.
It cannot be instantiated just like a abstract class.

Why use Java interface?

There are mainly three reasons to use interface. They are given below.

- It is used to achieve abstraction.

Date :

- By interface, we can support the functionality of multiple inheritance.
- It can be used to achieve loose coupling.

Interface Syntax:

```
interface <interface_name>{
  // declare constant fields
  // declare methods that abstract
  // by default.
}
```

Java interface Example

```
interface printable{
 void print();
}
class A6 implements printable{
 public void print(){ system.out.println("Hello"); }

 public static void main(string args[]){
   A6 obj = new A6();
   obj.print();
  }
}
```

Output:

```
Hello
```

Date :

8.3 Difference between abstract class and interface:

Abstract class	Interface
Abstract class have abstract and non-abstract method.	Interface can have only abstract methods.
Doesn't support multiple Inheritance.	Supports Multiple inheritance.
Can provide the implementation. of interface.	Can't provide implementation of abstract class.
Abstract class can have class members like private, protected, etc	Members of java interface are public by default.